

2024, Mar. 2024 Release LOC:Acute Adult**Infection: Pyelonephritis or Complex UTI**Utilization Benchmarks: *Multiple options available, see table below for options***Overview****Select Day****Initial review** ^(1, 2, 3, 4)**Episode Day 1** ^(1, 3, 4)**Episode Day 2** ^(3, 4, 62)**Episode Day 3** ^(3, 4, 82)**Episode Day 4****Discharge Screens** ⁽⁸⁴⁾**Utilization Benchmarks****Notes**

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Overview**Introduction:**

Urinary tract infection (UTI) is a broad term that includes infection of the bladder (cystitis), urethra (urethritis), ureters, or kidneys (pyelonephritis). Patients with cystitis are almost exclusively managed in the outpatient setting with oral anti-infectives; systemic symptoms of infection are unlikely to develop in patients with cystitis. UTI may be considered complicated or uncomplicated. Complicated infections are those that relate to structural or functional abnormalities or in an immunocompromised state. Uncomplicated infections are those present in otherwise healthy patients, with a structurally and functionally normal urinary tract (Ron Walls, In: Rosen's Emergency Medicine: Concepts And Clinical Practice, 9th edn. 2018; Cooper et al., Campell-Walsh-Wein Urology, Twelfth edition. ed. 2021). Application of criteria within this subset are intended to capture patients with both complicated and uncomplicated infection that result in systemic findings (including those with septic features) necessitating hospital level services across all levels of care.

Evaluation and Treatment:

The diagnosis of complicated urinary tract infection or pyelonephritis is confirmed with a positive urinalysis and urine culture, along with clinical findings. The use of routine blood cultures is not recommended. Blood cultures should be obtained when the patient has an uncertain diagnosis, is immunocompromised, or has a suspected hematogenous infection (Johnson and Russo, N Engl J Med 2018, 378: 48-59). In the pregnant patient, blood cultures are recommended if the patient does not respond to treatment and continues to have symptoms such as a fever of 102.2°F(39.0° C) or greater, signs of sepsis, or a significant complication such as acute respiratory distress syndrome (Gomi et al., Cochrane Database Syst Rev 2015: CD009216).

Patients who require hospitalization should receive an intravenous anti-infective. Most patients respond to treatment within 72 hours. If there is no improvement, a change to the anti-infective regimen and imaging are recommended. Renal ultrasound, although useful in determining if the patient has abnormal urinary tract anatomy without a radiation risk, is not as accurate as computed tomography (CT) of the pelvis and abdomen in diagnosing pyelonephritis (American College of Radiology, ACR appropriateness criteria: acute pyelonephritis. 2012. Revised 2018.).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were

assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2, 3, 4)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, All:** ⁽⁵⁾● Urinary symptoms and abnormal urinalysis and, ≥ **One:** ^(6, 7)– ≥ 10 WBC/mm³ ⁽⁸⁾– Leukocyte esterase positive ⁽⁹⁾– Nitrite positive ⁽¹⁰⁾

– Urine culture pending

● Finding persistent after treatment (excludes prehospital), ≥ **One:** ⁽¹¹⁾– Temperature ≥ 100.4°F(38.0°C) persistent after antipyretic (includes PO) ⁽¹²⁾– Severe pain unresponsive to ≥ 2 doses of analgesic prior to admit decision ⁽¹³⁾– Vomiting, persistent and unresponsive to ≥ 2 doses antiemetic ⁽¹³⁾● **ACUTE, All:** ⁽¹⁴⁾● Urinary symptoms with abnormal urinalysis and, ≥ **One:** ^(6, 7)– ≥ 10 WBC/mm³ ⁽⁸⁾– Leukocyte esterase positive ⁽⁹⁾– Nitrite positive ⁽¹⁰⁾● Systemic infection or end organ compromise, ≥ **One:**– Blood cultures positive ^(15, 16)– Pregnancy and ≥ 24 wks gestation ⁽¹⁷⁾– Urinary stent by history ⁽¹⁸⁾– Urinary tract obstruction or anatomical abnormality by imaging or history ⁽¹⁹⁾● Organ dysfunction or hypoperfusion, ≥ **One:** ⁽²⁰⁾– Creatinine > 1.5x baseline and < 2x baseline or estimated baseline ^(21, 22)– Lactic acid > ULN and < 4 mmol/L(36.0 mg/dL) ⁽²³⁾● Moderate risk finding, ≥ **Two:**– Mental status changes (excludes coma, stupor, or obtundation) and GCS 13-14 ⁽²⁴⁾– Platelet count 100,000/cu.mm(100x10⁹/L) to 150,000/cu.mm(150x10⁹/L) and < baseline ⁽²⁵⁾– Total bilirubin 1.2 mg/dL(20.52 μmol/L) to 1.9 mg/dL(32.49 μmol/L) and > baseline ⁽²⁵⁾– O₂ sat ≤ 91%(0.91) and < baseline ⁽²⁵⁾● High risk finding, ≥ **One:**– Mental status changes (excludes coma, stupor, or obtundation) and GCS 10-12 ⁽²⁴⁾– Platelet count < 100,000/cu.mm(100x10⁹/L) and < baseline ⁽²⁵⁾– Total bilirubin ≥ 2 mg/dL(34.2 μmol/L) and > baseline ⁽²⁵⁾● High risk for disease progression, **Both:**● Risk factor, ≥ **One:** ^(26, 27)

– Age > 75 years

● Coexisting disease, ≥ **One:** ⁽²⁸⁾– COPD, Grade 3 or 4 ⁽²⁹⁾

– Chronic kidney disease

– Diabetes mellitus

– Heart failure, NYHA Class III or IV ⁽³⁰⁾

– Liver disease

– Malignancy requiring active treatment with chemotherapy or radiation

– Immunocompromised ⁽³¹⁾● Finding, ≥ **One:**– Mental status changes (excludes coma, stupor, or obtundation) and GCS 13-14 ⁽²⁴⁾– Platelet count 100,000/cu.mm(100x10⁹/L) to 150,000/cu.mm(150x10⁹/L) and < baseline ⁽²⁵⁾– Total bilirubin 1.2 mg/dL(20.52 μmol/L) to 1.9 mg/dL(32.49 μmol/L) and > baseline ⁽²⁵⁾– O₂ sat ≤ 91%(0.91) and < baseline ⁽²⁵⁾

- **SIRS, ≥ Two:** ⁽³²⁾
 - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽¹²⁾
- **WBC, ≥ One:**
 - > 12,000/cu.mm(12x10⁹/L) or < 4,000/cu.mm(4x10⁹/L)
 - Bands > 10%(0.10)
- Heart rate > 90/min, sustained ⁽³³⁾
- Respiratory rate > 20/min, sustained ⁽³³⁾
- **INTERMEDIATE, One:** ⁽³⁴⁾
 - Sepsis, actual or suspected and hemodynamic stability, **Both:** ^(35, 36)
 - **Finding, ≥ One:**
 - **SIRS, ≥ Two:** ⁽³²⁾
 - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽¹²⁾
 - **WBC, ≥ One:**
 - > 12,000/cu.mm(12x10⁹/L) or < 4,000/cu.mm(4x10⁹/L)
 - Bands > 10%(0.10)
 - Heart rate > 90/min, sustained ⁽³³⁾
 - Respiratory rate > 20/min, sustained ⁽³³⁾
 - **High risk for mortality, Both:**
 - **High risk, ≥ One:**
 - **Quick Sequential [Sepsis-Related] Organ Failure Assessment (qSOFA), ≥ Two:** ⁽³⁷⁾
 - Mental status changes (excludes coma, stupor, or obtundation) and GCS 10-14 ⁽²⁴⁾
 - Respiratory rate ≥ 22/min, sustained ⁽³³⁾
 - Systolic blood pressure 90-100 mmHg
 - Lactic acid ≥ 4 mmol/L(36.0 mg/dL) ^(38, 39)
 - **Finding, ≥ One:**
 - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽¹²⁾
 - **WBC, ≥ One:** ⁽⁴⁰⁾
 - > 12,000/cu.mm(12x10⁹/L) or < 4,000/cu.mm(4x10⁹/L)
 - Bands > 10%(0.10)
 - Heart rate > 90/min, sustained ⁽³³⁾
- **Intervention, ≥ One:**
 - NIPPV ⁽⁴¹⁾
 - Requiring oxygen ≥ 40%(0.40), ≤ 2d ⁽⁴²⁾
 - **Volume expander, ≥ One:** ^(43, 44)
 - Crystalloid ≥ 30ml/kg/24h or colloid equivalent
 - Crystalloid ≥ 2L/24h or colloid equivalent
- **CRITICAL, One:** ⁽⁴⁵⁾
 - **Sepsis, actual or suspected, ≥ One:**
 - ARDS ⁽⁴⁶⁾
 - Coma, stupor, obtundation, or GCS ≤ 9 ⁽⁴⁷⁾
 - CRRT ⁽⁴⁸⁾
 - DIC and blood product transfusion ^(49, 50)
 - ECMO or ECLS ⁽⁵¹⁾
 - **Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, Both:** ⁽⁵²⁾
 - **Hypotension, ≥ One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - **Finding, ≥ One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽⁵³⁾
 - Urine output < 0.5 mL/kg/h
 - Hemodynamic monitoring, invasive ⁽⁵⁴⁾
 - HFNC and respiratory monitoring every 1-2h ^(42, 55, 56)

- Mechanical ventilation
- NIPPV and, ≥ **One:** ⁽⁴¹⁾
 - Arterial $\text{Po}_2 \leq 39$ mmHg(5.2 kPa)
 - Arterial or venous $\text{Pco}_2 \geq 60$ mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous $\text{pH} \leq 7.24$
 - No hx of COPD
 - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure) ^(57, 58)
- **Episode Day 1, One:** ^(1, 3, 4)
(Symptom or finding within 24h)
- **OBSERVATION, All:** ⁽⁵⁾
 - Urinary symptoms and abnormal urinalysis and, ≥ **One:** ^(6, 7)
 - ≥ 10 WBC/mm³ ⁽⁸⁾
 - Leukocyte esterase positive ⁽⁹⁾
 - Nitrite positive ⁽¹⁰⁾
 - Urine culture pending
 - Finding persistent after treatment (excludes prehospital), ≥ **One:** ⁽¹¹⁾
 - Temperature $\geq 100.4^\circ\text{F}(38.0^\circ\text{C})$ persistent after antipyretic (includes PO) ⁽¹²⁾
 - Severe pain unresponsive to ≥ 2 doses of analgesic prior to admit decision ⁽¹³⁾
 - Vomiting, persistent and unresponsive to ≥ 2 doses antiemetic ⁽¹³⁾
 - Anti-infective and, ≥ **One:** ⁽⁵⁹⁾
 - IV fluid ⁽⁶⁰⁾
 - Antiemetic $\geq 3\text{x}/24\text{h}$ or serotonin antagonist $\geq 2\text{x}/24\text{h}$
 - Requiring additional analgesic ≥ 2 doses
- **ACUTE, All:** ⁽¹⁴⁾
 - Urinary symptoms with abnormal urinalysis and, ≥ **One:** ^(6, 7)
 - ≥ 10 WBC/mm³ ⁽⁸⁾
 - Leukocyte esterase positive ⁽⁹⁾
 - Nitrite positive ⁽¹⁰⁾
 - Systemic infection or end organ compromise, ≥ **One:**
 - Blood cultures positive ^(15, 16)
 - Pregnancy and ≥ 24 wks gestation ⁽¹⁷⁾
 - Urinary stent by history ⁽¹⁸⁾
 - Urinary tract obstruction or anatomical abnormality by imaging or history ⁽¹⁹⁾
 - Organ dysfunction or hypoperfusion, ≥ **One:** ⁽²⁰⁾
 - Creatinine $> 1.5\text{x}$ baseline and $< 2\text{x}$ baseline or estimated baseline ^(21, 22)
 - Lactic acid $> \text{ULN}$ and < 4 mmol/L(36.0 mg/dL) ⁽²³⁾
 - Moderate risk finding, ≥ **Two:**
 - Mental status changes (excludes coma, stupor, or obtundation) and GCS 13-14 ⁽²⁴⁾
 - Platelet count 100,000/cu.mm($100 \times 10^9/\text{L}$) to 150,000/cu.mm($150 \times 10^9/\text{L}$) and $< \text{baseline}$ ⁽²⁵⁾
 - Total bilirubin 1.2 mg/dL($20.52 \mu\text{mol/L}$) to 1.9 mg/dL($32.49 \mu\text{mol/L}$) and $> \text{baseline}$ ⁽²⁵⁾
 - O_2 sat $\leq 91\%(0.91)$ and $< \text{baseline}$ ⁽²⁵⁾
 - High risk finding, ≥ **One:**
 - Mental status changes (excludes coma, stupor, or obtundation) and GCS 10-12 ⁽²⁴⁾
 - Platelet count $< 100,000/\text{cu.mm}(100 \times 10^9/\text{L})$ and $< \text{baseline}$ ⁽²⁵⁾
 - Total bilirubin ≥ 2 mg/dL($34.2 \mu\text{mol/L}$) and $> \text{baseline}$ ⁽²⁵⁾
 - High risk for disease progression, **Both:**
 - Risk factor, ≥ **One:** ^(26, 27)
 - Age > 75 years
 - Coexisting disease, ≥ **One:** ⁽²⁸⁾
 - COPD, Grade 3 or 4 ⁽²⁹⁾
 - Chronic kidney disease
 - Diabetes mellitus
 - Heart failure, NYHA Class III or IV ⁽³⁰⁾

- Liver disease
 - Malignancy requiring active treatment with chemotherapy or radiation
 - Immunocompromised ⁽³¹⁾
 - Finding, ≥ **One**:
 - Mental status changes (excludes coma, stupor, or obtundation) and GCS 13-14 ⁽²⁴⁾
 - Platelet count 100,000/cu.mm($100 \times 10^9/L$) to 150,000/cu.mm($150 \times 10^9/L$) and < baseline ⁽²⁵⁾
 - Total bilirubin 1.2 mg/dL($20.52 \mu\text{mol/L}$) to 1.9 mg/dL($32.49 \mu\text{mol/L}$) and > baseline ⁽²⁵⁾
 - O_2 sat ≤ 91%(0.91) and < baseline ⁽²⁵⁾
 - SIRS, ≥ **Two**: ⁽³²⁾
 - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽¹²⁾
 - WBC, ≥ **One**:
 - > 12,000/cu.mm($12 \times 10^9/L$) or < 4,000/cu.mm($4 \times 10^9/L$)
 - Bands > 10%(0.10)
 - Heart rate > 90/min, sustained ⁽³³⁾
 - Respiratory rate > 20/min, sustained ⁽³³⁾
 - Urine culture pending and, **Both**:
 - Anti-infective ^(59, 61)
 - IV fluid ⁽⁶⁰⁾
 - INTERMEDIATE, **One**: ⁽³⁴⁾
 - Sepsis, actual or suspected and hemodynamic stability, **Both**: ^(35, 36)
 - Finding, ≥ **One**:
 - SIRS, ≥ **Two**: ⁽³²⁾
 - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽¹²⁾
 - WBC, ≥ **One**:
 - > 12,000/cu.mm($12 \times 10^9/L$) or < 4,000/cu.mm($4 \times 10^9/L$)
 - Bands > 10%(0.10)
 - Heart rate > 90/min, sustained ⁽³³⁾
 - Respiratory rate > 20/min, sustained ⁽³³⁾
 - High risk for mortality, **Both**:
 - High risk, ≥ **One**:
 - Quick Sequential [Sepsis-Related] Organ Failure Assessment (qSOFA), ≥ **Two**: ⁽³⁷⁾
 - Mental status changes (excludes coma, stupor, or obtundation) and GCS 10-14 ⁽²⁴⁾
 - Respiratory rate ≥ 22/min, sustained ⁽³³⁾
 - Systolic blood pressure 90-100 mmHg
 - Lactic acid ≥ 4 mmol/L(36.0 mg/dL) ^(38, 39)
 - Finding, ≥ **One**:
 - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽¹²⁾
 - WBC, ≥ **One**: ⁽⁴⁰⁾
 - > 12,000/cu.mm($12 \times 10^9/L$) or < 4,000/cu.mm($4 \times 10^9/L$)
 - Bands > 10%(0.10)
 - Heart rate > 90/min, sustained ⁽³³⁾
 - Intervention, ≥ **One**:
 - NIPPV ⁽⁴¹⁾
 - Requiring oxygen ≥ 40%(0.40), ≤ 2d ⁽⁴²⁾
 - Volume expander, ≥ **One**: ^(43, 44)
 - Crystalloid ≥ 30ml/kg/24h or colloid equivalent
 - Crystalloid ≥ 2L/24h or colloid equivalent
- CRITICAL, **One**: ⁽⁴⁵⁾
 - Sepsis, actual or suspected, ≥ **One**:
 - ARDS ⁽⁴⁶⁾
 - Coma, stupor, obtundation, or GCS ≤ 9 ⁽⁴⁷⁾
 - CRRT ⁽⁴⁸⁾
 - DIC and blood product transfusion ^(49, 50)

- ECMO or ECLS ⁽⁵¹⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24 h and, **Both:** ⁽⁵²⁾
 - Hypotension, $\geq$ **One:**
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and $<$ baseline
 - Finding, $\geq$ **One:**
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽⁵³⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽⁵⁴⁾
- HFNC and respiratory monitoring every 1-2h ^(42, 55, 56)
- Mechanical ventilation
- NIPPV and, $\geq$ **One:** ⁽⁴¹⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
- Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure) ^(57, 58)
- **Episode Day 2, One:** ^(3, 4, 62)
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽⁶³⁾
 - Afebrile or temperature within acceptable limits
 - Anti-infective, **One:**
 - Arranged
 - Completed
 - Not indicated
 - Laboratory values within acceptable limits ⁽⁶⁴⁾
 - Pain absent or within acceptable limits
 - Tolerating PO ⁽⁶⁵⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
 - Pyelonephritis or complex UTI unresolved and, **All:** ⁽⁶⁶⁾
 - Urine culture pending or positive
 - Finding, $\geq$ **One:**
 - Systemic infection, $\geq$ **One:**
 - Temperature $\geq 100.4^{\circ}\text{F}(38.0^{\circ}\text{C})$ or $< 97.0^{\circ}\text{F}(36.1^{\circ}\text{C})$ ⁽¹²⁾
 - WBC, $\geq$ **One:**
 - $> 12,000/\text{cu.mm}(12 \times 10^9/\text{L})$ or $< 4,000/\text{cu.mm}(4 \times 10^9/\text{L})$
 - Bands $> 10\%(0.10)$
 - Symptoms persistent, $\geq$ **One:**
 - Severe pain unresponsive to ≥ 2 doses of analgesic
 - Vomiting unresponsive to ≥ 2 doses antiemetic
 - Anti-infective or transition to PO ≤ 24 h and, $\geq$ **One:** ^(59, 61, 67)
 - Antiemetic $\geq 3 \times 24$ h or serotonin antagonist $\geq 2 \times 24$ h
 - IV fluid ⁽⁶⁰⁾
 - Requiring additional analgesic ≥ 2 doses
- **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ^(63, 68)
 - Afebrile or temperature within acceptable limits
 - Anti-infective, **One:**
 - Arranged
 - Completed
 - Not indicated
 - Laboratory values within acceptable limits ⁽⁶⁴⁾

- Pain absent or within acceptable limits
- Tolerating PO ⁽⁶⁵⁾
- **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁶⁹⁾
- **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - **Acute kidney injury (AKI), ≥ One:**
 - Kidney function improving or within acceptable limits ⁽⁶⁴⁾
 - Outpatient dialysis regimen established
 - Adverse drug reaction resolving
 - **Renal abscess resolving and, One:**
 - Percutaneous drain discontinued or not indicated
 - **Percutaneous drain in place and, One:**
 - Patient or caregiver able to safely manage care
 - Home care services arranged
- **Partial responder, not clinically stable for discharge and requires continued stay, ≥ One:**
 - **Pyelonephritis or complex UTI unresolved and, All: ⁽⁶⁶⁾**
 - Urine culture pending or positive
 - **Finding, ≥ One:**
 - Blood cultures positive ^(15, 16)
 - Severe pain requiring analgesic ≥ 2 doses
 - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽¹²⁾
 - **WBC, ≥ One:**
 - > 12,000/cu.mm($12 \times 10^9/L$) or < 4,000/cu.mm($4 \times 10^9/L$)
 - Bands > 10%(0.10)
 - ANC < 500/cu.mm($500 \times 10^6/L$) ⁽⁷⁰⁾
 - Anti-infective and IV fluid ^(59, 60)
- **AKI, Both: ⁽⁷¹⁾**
 - **Finding, ≥ One:**
 - Oliguria ^(72, 73)
 - **Creatinine, One: ⁽²²⁾**
 - ≥ 1.5x baseline and chronic kidney disease
 - ≥ 1.5x ULN ⁽⁷⁴⁾
 - GFR > 25%(0.25) decrease from baseline ⁽²⁵⁾
 - Volume expander ≤ 2d ^(43, 75)
- Adverse drug reaction requiring change in medication ≤ 24h ⁽⁷⁶⁾
- **Renal abscess confirmed by imaging and, All: ⁽⁷⁷⁾**
 - **Finding, ≥ One:**
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹²⁾
 - WBC ≥ 12,000/cu.mm($12 \times 10^9/L$)
 - Bands > 10%(0.10)
 - Anti-infective
 - **Drainage, ≥ One:**
 - Performed within last 24h
 - Persistent drainage
 - Requiring drain repositioning within last 24h
- Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - HFNC and respiratory monitoring every 3-4h ^(55, 56)
 - NIPPV ⁽⁴¹⁾
 - Oxygen ≥ 40%(0.40), ≤ 2d ⁽⁴²⁾
- **CRITICAL, ≥ One:**
 - **Acute kidney injury (AKI) and, Both: ⁽⁷¹⁾**
 - **Finding, ≥ One:**

- Potassium > 6.0 mEq/L(6.0 mmol/L) and, ≥ **One**:
 - Loss of P wave
 - Multifocal PVCs
 - Neuromuscular deficit ⁽⁷⁸⁾
 - Widening QRS
- Sodium, **One**:
 - < 120 mEq/L(120 mmol/L) and volume overload
 - > 160 mEq/L(160 mmol/L)
- Intervention, ≥ **One**:
 - Calcium chloride
 - Calcium gluconate
 - Glucose 50%(0.50) with insulin ⁽⁷⁹⁾
 - Diuretic, continuous
- Dialysis, **One**:
 - Hemodialysis ≤ 7d since initiation
 - Peritoneal dialysis ≤ 7d since initiation
 - IV fluid resuscitation ≤ 2d ⁽⁸⁰⁾
- CRRT ⁽⁴⁸⁾
- DIC ⁽⁴⁹⁾
- ECMO or ECLS ⁽⁵¹⁾
- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation ≤ 24h and, **Both**: ⁽⁵²⁾
 - Hypotension, ≥ **One**:
 - MAP < 65 mmHg
 - Systolic blood pressure < 90 mmHg and < baseline
 - Finding, ≥ **One**:
 - Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽⁵³⁾
 - Urine output < 0.5 mL/kg/h
- Hemodynamic monitoring, invasive ⁽⁵⁴⁾
- HFNC and respiratory monitoring every 1-2h ^(42, 55, 56)
- Mechanical ventilation
- NIPPV and, ≥ **One**: ⁽⁴¹⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁸¹⁾
- ⁵⁸⁾Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure) ^{(57,}
- **Episode Day 3, One**: ^(3, 4, 82)
 - **OBSERVATION, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽⁶³⁾
 - Afebrile or temperature within acceptable limits
 - **Anti-infective, One**:
 - Arranged
 - Completed
 - Not indicated
 - Laboratory values within acceptable limits ⁽⁶⁴⁾
 - Pain absent or within acceptable limits
 - Tolerating PO ⁽⁶⁵⁾
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One**:
 - For pyelonephritis or complex UTI use Episode Day 3 at a higher level of care
 - For a condition other than pyelonephritis or complex UTI use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review

● **ACUTE, One:**● **Responder**, discharge expected today if clinically stable last 12h, **All:** ^(63, 68)

- Afebrile or temperature within acceptable limits

● **Anti-infective, One:**

- Arranged
- Completed
- Not indicated
- Laboratory values within acceptable limits ⁽⁶⁴⁾
- Pain absent or within acceptable limits
- Tolerating PO ⁽⁶⁵⁾

● **Complication or comorbidity, One:**

- No complication or active comorbidity relevant to this episode of care ⁽⁶⁹⁾

● **Complication or comorbidity and clinically stable for discharge, ≥ One:**● **Acute kidney injury (AKI), ≥ One:**

- Kidney function improving or within acceptable limits ⁽⁶⁴⁾
- Outpatient dialysis regimen established
- Adverse drug reaction resolving

● **Renal abscess resolving and, One:**

- Percutaneous drain discontinued or not indicated

● **Percutaneous drain in place and, One:**

- Patient or caregiver able to safely manage care
- Home care services arranged

● **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**● **Pyelonephritis or complex UTI unresolved and, All:** ⁽⁶⁶⁾

- Urine culture positive

● **Finding, ≥ One:**

- Blood cultures positive ^(15, 16)
- Positive multidrug-resistant bacteria (excludes colonization) ⁽⁸³⁾
- Severe pain requiring analgesic ≥ 2 doses
- Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) ⁽¹²⁾

● **WBC, ≥ One:**

- > 12,000/cu.mm(12x10⁹/L) or < 4,000/cu.mm(4x10⁹/L)
- Bands > 10%(0.10)
- ANC < 500/cu.mm(500x10⁶/L) ⁽⁷⁰⁾

- Anti-infective ^(59, 61, 67)

● **AKI, Both:** ⁽⁷¹⁾● **Finding, ≥ One:**

- Oliguria ^(72, 73)

● **Creatinine, One:** ⁽²²⁾

- ≥ 1.5x baseline and chronic kidney disease
- ≥ 1.5x ULN ⁽⁷⁴⁾

- GFR > 25%(0.25) decrease from baseline ⁽²⁵⁾

- Volume expander ≤ 2d ^(43, 75)

- Adverse drug reaction requiring change in medication ≤ 24h ⁽⁷⁶⁾

● **Renal abscess confirmed by imaging and, All:** ⁽⁷⁷⁾● **Finding, ≥ One:**

- Temperature ≥ 100.4°F(38.0°C) ⁽¹²⁾
- WBC ≥ 12,000/cu.mm(12x10⁹/L)
- Bands > 10%(0.10)

- Anti-infective ⁽⁵⁹⁾

● **Drainage, ≥ One:**

- Performed within last 24h
- Persistent drainage

- Requiring drain repositioning within last 24h
- Post Critical care $\leq 24h$

● **INTERMEDIATE, $\geq$ One:**

- HFNC and respiratory monitoring every 3-4h ^(55, 56)
- NIPPV ⁽⁴¹⁾
- Oxygen $\geq 40\%(0.40)$, $\leq 2d$ ⁽⁴²⁾

● **CRITICAL, $\geq$ One:**

- Acute kidney injury (AKI) and, **Both:** ⁽⁷¹⁾

● Finding, $\geq$ One:

- Potassium > 6.0 mEq/L(6.0 mmol/L) and, $\geq$ One:

- Loss of P wave
- Multifocal PVCs
- Neuromuscular deficit ⁽⁷⁸⁾
- Widening QRS

● Sodium, **One:**

- < 120 mEq/L(120 mmol/L) and volume overload
- > 160 mEq/L(160 mmol/L)

● Intervention, $\geq$ One:

- Calcium chloride
- Calcium gluconate
- Glucose 50%(0.50) with insulin ⁽⁷⁹⁾
- Diuretic, continuous

● Dialysis, **One:**

- Hemodialysis $\leq 7d$ since initiation
- Peritoneal dialysis $\leq 7d$ since initiation
- IV fluid resuscitation $\leq 2d$ ⁽⁸⁰⁾

- CRRT ⁽⁴⁸⁾

- DIC ⁽⁴⁹⁾

- ECMO or ECLS ⁽⁵¹⁾

- Hemodynamic instability unresponsive to ≥ 2 liter fluid bolus, requiring ongoing fluid resuscitation $\leq 24h$ and, **Both:** ⁽⁵²⁾

● Hypotension, $\geq$ One:

- MAP < 65 mmHg
- Systolic blood pressure < 90 mmHg and $<$ baseline

● Finding, $\geq$ One:

- Lactic acid > 2 mmol/L(18.0 mg/dL) ⁽⁵³⁾
- Urine output < 0.5 mL/kg/h

- Hemodynamic monitoring, invasive ⁽⁵⁴⁾

- HFNC and respiratory monitoring every 1-2h ^(42, 55, 56)

- Mechanical ventilation

- NIPPV and, $\geq$ One: ⁽⁴¹⁾

- FiO₂ $> 50\%(0.50)$
- Respiratory intervention every 1-2h ⁽⁸¹⁾

- Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure) ^(57, 58)

● **Episode Day 4, One:**

● **ACUTE, One:**

- **Responder**, discharge expected today if clinically stable last 12h, **All:** ^(63, 68)

- Afebrile or temperature within acceptable limits

● **Anti-infective, One:**

- Arranged
- Completed
- Not indicated

- Laboratory values within acceptable limits ⁽⁶⁴⁾
- Pain absent or within acceptable limits
- Tolerating PO ⁽⁶⁵⁾
- **Complication or comorbidity, One:**
 - No complication or active comorbidity relevant to this episode of care ⁽⁶⁹⁾
- **Complication or comorbidity and clinically stable for discharge, ≥ One:**
 - **Acute kidney injury (AKI), ≥ One:**
 - Kidney function improving or within acceptable limits ⁽⁶⁴⁾
 - Outpatient dialysis regimen established
 - Adverse drug reaction resolving
 - **Renal abscess resolving and, One:**
 - Percutaneous drain discontinued or not indicated
 - **Percutaneous drain in place and, One:**
 - Patient or caregiver able to safely manage care
 - Home care services arranged
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
- **INTERMEDIATE, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review
- **CRITICAL, One:**
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
 - **Use Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, ≥ One: ⁽⁸⁴⁾

- **HOME, Both:**
 - **Level of care appropriateness, Both:**
 - Home environment safe and accessible ⁽⁸⁵⁾
 - Patient or caregiver demonstrates ability to manage care
- **Pyelonephritis or complex UTI discharge planning, All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁸⁶⁾
 - **Education on the importance of, All:**
 - Understanding the signs of UTI and when to seek treatment
 - Prevention
 - Completing full antibiotic course
 - Medication reconciliation ⁽⁸⁷⁾
 - Follow-up care planned ⁽⁸⁸⁾
 - Identify and address transportation needs
- **HOME CARE, Both:**
 - **Level of care appropriateness, All:**
 - Home environment safe and accessible ⁽⁸⁵⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(89, 90)
 - **Skilled services, ≥ One:** ⁽⁹¹⁾
 - Skilled nursing ⁽⁹²⁾
 - Skilled therapy, PT, OT, or ST
 - Behavioral health ⁽⁹³⁾

- Discharge planning, **All:**
 - Provide comprehensive written discharge and teaching instructions ⁽⁸⁶⁾
- Education on the importance of, **All:**
 - Understanding the signs of UTI and when to seek treatment
 - Prevention
 - Completing full antibiotic course
- Medication reconciliation ⁽⁸⁷⁾
- Follow-up care planned ⁽⁸⁸⁾
- Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
 - Level of care appropriateness, **One:**
 - Support systems available ⁽⁹⁴⁾
 - Skilled treatment, **≥ One:**
 - Clinical assessment ⁽⁹⁵⁾
 - Individual, group, or family counseling
 - Psychiatric, substance, or medication evaluation ⁽⁹⁶⁾
- **SKILLED MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 1\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁹²⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(97, 98)
 - Functional impairments requiring at least supervision ⁽⁹⁹⁾
 - Able to tolerate 1h/d of skilled therapy $\geq 5\text{d/wk}$
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 2\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **≥ One:**
 - Medical and skilled nursing interventions or assessment 4h/24h ⁽⁹²⁾
 - Therapy, **All:**
 - Goal directed therapy with rehabilitation potential ^(97, 98)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹⁹⁾
 - Able to tolerate 2-3h/d of skilled therapy $\geq 5\text{d/wk}$ and ≥ 1 therapy discipline
 - Complete prior to facility transfer, **All:**
 - Medication reconciliation ⁽⁸⁷⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation
- **SUBACUTE REHAB, Both:**
 - Level of care appropriateness, **All:**
 - Medical practitioner, NP, PA assessment or oversight $\geq 3\text{x/wk}$
 - Treatment precluded in a lower level
 - Services, **All:**
 - Goal directed therapy with rehabilitation potential ^(97, 98)
 - ≥ 2 functional impairments requiring at least minimum assistance ⁽⁹⁹⁾

– Able to tolerate 2-3h/d of skilled therapy $\geq$ 5d/wk and $\geq$ 2 therapy disciplines

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁸⁷⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **ACUTE REHAB, Both:**

● Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight $\geq$ 3x/wk

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(97, 98)
- $\geq$ 2 functional impairments requiring at least minimum assistance ⁽⁹⁹⁾
- Able to tolerate $\geq$ 15h/wk of skilled therapy and $\geq$ 2 therapy disciplines ⁽¹⁰⁰⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁸⁷⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

● Level of care appropriateness, **Both:**

- Treatment precluded in a lower level

● In lieu of acute or continued hospitalization or failed lower level of care, $\geq$ **One:**

- Primary condition or illness and treatment of active comorbid condition(s) ⁽¹⁰¹⁾
- Ventilator dependent $\geq$ 6h/d and weaning planned ^(102, 103)
- Acute and comorbid conditions requiring prolonged hospitalization ⁽¹⁰⁴⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁸⁷⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
689 KIDNEY AND URINARY TRACT INFECTIONS WITH MCC	n/a	3.9	CMS GMLOS
690 KIDNEY AND URINARY TRACT INFECTIONS WITHOUT MCC	n/a	2.9	CMS GMLOS
PYELONEPHRITIS	31%	3	InterQual
URINARY TRACT INFECTION, COMPLEX	35%	3.8	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
 - Unexpected death
 - Transfer of the patient
 - Unexpected clinical improvement
 - Election of hospice care
-
- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
 - Surgical procedure or intervention found on the CMS Inpatient Only List
-
- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Expected progress within 24-48h:

- Vital signs improving
- Laboratory values normalizing
- Activity increasing
- Tolerating PO
- Pain minimal or pain free

Consider:

- Changing antibiotic route from parenteral to PO
- Discharge planning

4:

Care facilitation:

- Potential risk of readmission
- Knowledge deficit or non-adherence with condition management identified (**Home Care, Readmission reduction team**)
- Co-morbid chronic obstructive pulmonary disease or heart failure (**Home Care, Readmission reduction team**)

Functional or ADL impairment - consider an alternate level of care for therapy needs

- Physical therapy provided in the home setting should be considered if the home is safe and accessible (**Home Care**)
- If home environment is unsafe, considered a skilled nursing facility (**SNF**)

Social Determinants of Health (SDOH)

Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.

Alcohol or substance use

- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals

- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice

- Consider palliative care for patients with end stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited

- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, home homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or ADL impairment

- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved

- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission

- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence

- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance
- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers.

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies, i.e., blood lead level testing and follow up (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

5:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

6:

This criteria point is intended to exclude patients who are chronically colonized at baseline, presenting without urinary symptoms. Patients with altered mental status and no obvious urinary symptoms by history should be evaluated for costovertebral angle (CVA) or suprapubic tenderness on examination (Long and Koyfman, Emerg Med Clin North Am 2018, 36: 685-710). Patients not demonstrating one of the listed urinary symptoms may be appropriate for treatment of suspected urinary tract infection, but would require secondary review as this should be considered on a case by case basis. Asymptomatic bacteriuria is a common finding across all populations, with notable variations by age, gender, presence of genitourinary abnormality, or other factors.

7:

Urinary symptoms may include:

- Costovertebral angle (CVA) pain or tenderness
- Dysuria
- Frequency
- Flank pain
- Hematuria, gross
- Incontinence
- Suprapubic pain, pressure, or tenderness
- Urgency

8:

A WBC count greater than $10/\text{mm}^3$ in a urinalysis usually indicates a bacterial infection in the urinary tract (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23; Centers for Disease Control and Prevention, Urinary Tract Infection Events. 2015).

9:

A positive leukocyte esterase on the urinalysis indicates WBCs in the urine (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23; Centers for Disease Control and Prevention, Urinary Tract Infection Events. 2015).

10:

Positive nitrites in the urinalysis are a highly sensitive (95-98%) indicator for urinary tract infection (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23).

11:

Observation level of care may be appropriate for patients with pyelonephritis or complicated urinary tract infection who require more extensive supportive care (e.g., IV fluids, antiemetics, or pain management) and initial IV antibiotic treatment. This includes patients who are unable to tolerate oral antibiotics and it is expected that their symptoms will improve in 24 hours (Johnson and Russo, N Engl J Med 2018, 378: 48-59; Suri and Aurora, Emerg Med Clin North Am 2017, 35: 647-71).

12:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

13:

Indications for continued treatment and monitoring for patients with acute pyelonephritis or complex urinary tract infection include uncontrolled vomiting and other complicating factors, such as pain management (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23; Johnson and Russo, N Engl J Med 2018, 378: 48-59).

14:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

15:

This criteria point may be satisfied by:

- Blood culture growth within the first 24 hours acknowledged on a retrospective Episode Day 1 review (e.g., those drawn in the Emergency Department (ED))
- Blood culture growth from a specimen collected up to one week prior to hospital admission (e.g., those sent by an outpatient clinic, center, office, or prior ED visit)

This criteria point includes patients who exhibit growth on blood cultures collected on admission (usually in the ED), as well as patients with growth on blood cultures collected up to one week prior to hospital admission. For patients who develop signs and symptoms of a systemic infection while hospitalized for another reason, similar considerations for timing of blood cultures apply, and the criteria point can be satisfied by positive blood cultures that were drawn up to one week prior to development of a systemic infection. Patients presenting with signs and symptoms of a systemic infection of unknown source, any growth on blood cultures is expected to be initially treated as indicative of a bloodstream infection. Ultimate determination of whether positive blood cultures represent true bloodstream infection or a contaminant primarily rests on the identification of the organism, with other factors taken into consideration, including the final number of culture bottles and sets exhibiting growth, time to blood culture positivity, presence or absence of growth on repeat blood cultures, as well as the clinical context that led to the blood cultures being obtained (Dargere et al., Clin Microbiol Infect 2018, 24: 964-9). The first report of growth commonly includes a description of the gram stain of the bacterial organism or the presence of yeast (i.e., "gram positive cocci in pairs and chains in the aerobic bottle"), with the final identification of the organism often unavailable until 2 or more days after the blood cultures are obtained. A prospective study demonstrated that systemic inflammatory response syndrome (SIRS) criteria are predictive of true bloodstream infections in a cohort of inpatients with positive blood cultures (Leth et al., J Emerg Med 2013, 44: 550-7). Another cohort study demonstrated that presence of SIRS criteria in patients who had blood cultures drawn was associated with increased mortality (Lindvig et al., European Journal of Emergency Medicine 2016, 23:). A prospective study of over 35,000 patients showed that patients who had growth on blood cultures, which were drawn in the ED for suspected sepsis, had higher mortality and longer length of stay (Chotirmall et al., Eur J Emerg Med 2016, 23: 38-43).

16:

Blood cultures are useful in patients with suspected urinary tract infection and bacteremia to establish the diagnosis or uro-pathogen (e.g., in patients who have asymptomatic bacteriuria or previously received antibiotic therapy) and identify those patients at higher risk requiring admission (Rothe et al., PLoS One 2020, 15: e0240981; Johnson and Russo, N Engl J Med 2018, 378: 48-59).

17:

Pyelonephritis or complicated urinary tract infection in pregnancy may require hospitalization due to the associated dehydration, nausea and vomiting, and the potential for preterm labor. Also, there is the uncommon but serious risk of endotoxic shock and endotoxic damage to alveolar capillary membranes which leads to pulmonary edema and the potential for acute respiratory distress syndrome. Patients who present at 24 weeks gestation or greater are at higher risk for complications related to pyelonephritis or complicated urinary tract infection. Hospitalization is recommended for management of patients who have moderate to severe symptoms (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23; Wing et al., Am J Obstet Gynecol 2014, 210: 219 e1-6).

18:

Patients with urinary stents or other indwelling devices, such as nephrostomy tubes or permanent foley catheter are at increased risk for complicated urinary tract infection as they have a higher risk of treatment failure and sepsis (Kozyrakakis et al., J Endourol 2020, 34: 516-22; Petrosillo et al., Expert Rev Anti Infect Ther 2020, 18: 47-61).

19:

Patients with a urinary tract infection due to a urinary obstruction are at increased risk for sepsis. Examples of urinary tract obstruction include malignancy and renal calculi. Anatomical obstruction may be due to

hydronephrosis or urinary fistula (Kozyrakis et al., J Endourol 2020, 34: 516-22; Petrosillo et al., Expert Rev Anti Infect Ther 2020, 18: 47-61).

20:

The following categories of organ dysfunction are derived from the Sequential [Sepsis-Related] Organ Failure Assessment (SOFA) score, as proposed in the Third International Consensus Definition for Sepsis and Septic Shock (Sepsis-3), with the following modifications: acute kidney injury is defined as an increase in serum creatinine with a reference to baseline creatinine (and not absolute numeric values used in SOFA), and oxygen saturation is used in place of the PaO₂/FiO₂ (partial pressure of oxygen in arterial blood/fraction of inspired oxygen) ratio given the lack of universal availability of arterial blood gas measurements for patients at the Acute level of care (Singer et al., JAMA 2016, 315: 801-10; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii-138).

21:

When the baseline serum creatinine level is unknown, the Modification of Diet in Renal Disease (MDRD) equation can be used to estimate an expected creatinine value (assuming a normal baseline value for glomerular filtration rate (GFR) of 75ml/min per 1.73 meters squared). The following values demonstrate one method to estimate baseline serum creatinine (eSCr) (with the omission of the race variable) (Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii-138; Bellomo et al., Crit Care 2004, 8: R204-12):

Male sex assigned at birth:

- Age 20-24 - eSCr 1.3 mg/dL (115 µmol/l)
- Age 25-39 - eSCr 1.2 mg/dL (106 µmol/l)
- Age 40-65 - eSCr 1.1 mg/dL (97 µmol/l)
- Age > 65 - eSCr 1.0 mg/dL (88 µmol/l)

Female sex assigned at birth:

- Age 20-29 - eSCr 1.0 mg/dL (88 µmol/l)
- Age 30-54 - eSCr 0.9 mg/dL (80 µmol/l)
- Age > 54 - eSCr 0.8 mg/dL (71 µmol/l)

Both the MDRD and the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equations provide a calculation of estimated GFR using serum creatinine, sex assigned at birth, age, and race. Recognition of substantial diversity within population groups has led to multiple organizations no longer including race in the estimated GFR calculations. This approach was investigated by a joint task force of the National Kidney Foundation and the American Society of Nephrology, the result of which was a recommendation to implement the new CKD-EPI creatinine equation which does not use the race variable (Delgado et al., Am J Kidney Dis 2022, 79: 268-88 e1; Inker et al., N Engl J Med 2021, 385: 1737-49).

22:

Patients with a urinary tract infection and a creatinine level greater than 1.5 mg/dl may require hospital admission (Kozyrakis et al., J Endourol 2020, 34: 516-22; Petrosillo et al., Expert Rev Anti Infect Ther 2020, 18: 47-61; Kang et al., Am J Emerg Med 2013, 31: 1067-72).

23:

Several studies have shown that patients with moderately elevated lactic acid levels (greater than normal but under 4 mmol/L) and evidence of infection are at increased risk for poor outcome (Villar et al., Infect Dis (Auckl) 2019, 12: 1178633719862776; Oedorf et al., West J Emerg Med 2017, 18: 258-66; Rhee et al., JAMA 2017, 318: 1241-9; Puskarich et al., J Crit Care 2014, 29: 334-9; Shapiro et al., Ann Emerg Med 2005, 45: 524-8).

24:

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).

Instruction: Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

25:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

26:

Patients with systemic infections and risk factors for the development of sepsis may require inpatient hospitalization. The following factors are associated with an increased risk of sepsis (Kempker and Martin, Clin Chest Med 2016, 37: 165-79; National Institute for Health and Care Excellence, Sepsis: recognition, diagnosis and early management. 2016; Rhodes et al., Crit Care Med 2017, 45: 486-552; Mayr et al., Virulence 2014, 5: 4-11):

- > 75 years of age
- Underlying chronic disease
- Malnutrition
- Immunocompromised state

27:

Factors associated with mortality and/or severe sepsis in patients with pyelonephritis and complicated urinary tract infection include (Long and Koyfman, Emerg Med Clin North Am 2018, 36: 685-710; Ruiz-Mesa et al., Medicine (Baltimore) 2017, 96: e8371; Buonaiuto et al., BMC Infect Dis 2014, 14: 639):

- Age > 75 years
- Anatomic abnormalities
- Antibiotic failure
- Bacteremia
- Chronic renal failure
- Health care associated infection
- Immunosuppression
- Kidney abscess
- Obstructive uropathy
- Thrombopenia

28:

The following comorbidities are associated with an increased risk of developing sepsis in the presence of an infection (Kempker and Martin, Clin Chest Med 2016, 37: 165-79; National Institute for Health and Care Excellence, Sepsis: recognition, diagnosis and early management. 2016; Rhodes et al., Crit Care Med 2017, 45: 486-552; Mayr et al., Virulence 2014, 5: 4-11):

- Chronic obstructive pulmonary disease
- Chronic renal disease
- Heart failure
- Diabetes mellitus
- Liver disease
- Malignancy

29:

The severity of chronic obstructive pulmonary disease (COPD) airflow limitations is classified into 4 grades based on the patient's baseline forced expiratory volume (FEV₁) measurement post-bronchodilator administration as follows (Global Initiative for Chronic Obstructive Lung Disease (GOLD). Global Strategy for the Diagnosis, Management and Prevention of COPD. 2024):

- GOLD 1 - Mild: FEV₁ ≥ 80%(0.80) predicted
- GOLD 2 - Moderate: FEV₁ 50-79%(0.50-0.79) predicted
- GOLD 3 - Severe: FEV₁ 30-49%(0.30-0.49) predicted
- GOLD 4 - Very severe: FEV₁ < 30%(0.30) predicted

30:

New York Heart Association (NYHA) classification for heart failure is defined as follows (Yap et al., Clinical cardiology 2015, 38: 621-8; New York Heart Association, Diseases of the Heart and Blood Vessels. Nomenclature and Criteria for diagnosis. 1964):

- Class I - No limitation of physical activity. Ordinary physical activity does not cause undue fatigue, palpitation, dyspnea, or anginal pain.
- Class II - Slight limitation of physical activity. Comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnea, or anginal pain.
- Class III - Marked limitation of physical activity. Comfortable at rest. Less than ordinary physical activity causes fatigue, palpitation, dyspnea, or anginal pain.
- Class IV - Inability to carry on any physical activity without discomfort. Symptoms of cardiac insufficiency or of anginal syndrome may be present even at rest. If any physical activity is undertaken, discomfort is increased.

31:

Immunocompromised refers to the patient who has a decreased capacity to defend against invading organisms as a result of an impaired immune system due to several comorbidities. The principal manifestation of immunodeficiency is increased susceptibility to infection with unusual severity, frequency, and duration which differs from that of an individual who is immunocompetent who may be exposed to the same pathogen. Furthermore, individuals who are immunocompromised are increasingly prone to opportunistic infections. Immunodeficiency is classified as either primary (congenital), caused by inherited disorders of the immune system, or secondary (acquired), as a result of a disease process or as a result of treatment of other chronic diseases. Primary causes of immunodeficiency account for more than 200 genetic disorders that impact immune system function. Some examples of primary immunodeficiency include congenital immunodeficiency diseases (e.g., X-linked agammaglobulinemia), combined immunodeficiency disease (e.g., DiGeorge syndrome, Wiskott syndrome), and chronic granulomatous disease.

Examples of secondary immune deficiency include poorly controlled diabetes mellitus defined as HbA1c 9%(0.09) (drawn on admission or within the past three months), end-stage liver disease, hematopoietic malignancies, patients who have had allogeneic hematopoietic stem cell transplantation (HSCT), and patients receiving immunosuppressive therapy (e.g., chemotherapy, radiation, anti-rejection medications, or prolonged high steroid use). Patients with severe, prolonged neutropenia (e.g., absolute neutrophil count less than 500/cu.mm(500x106/L), HSCT, human immunodeficiency virus (HIV) with CD4 count less than 200/cu.mm(200x106/L), a history of splenectomy, splenic dysfunction due to sickle cell disease, and those who have received intense chemotherapy (e.g., childhood acute myelogenous leukemia) are at greatest risk of infection (Centers for Disease Control and Prevention, National Diabetes Statistics Report. 2022; Bonilla et al., J Allergy Clin Immunol 2015, 136: 1186-205 e78; Lipska et al., Diabetes Care 2013, 36: 3535-42).

32:

Systemic inflammatory response syndrome (SIRS) is an inflammatory process defined as 2 or more of the following clinical findings: fever, leukocytosis, tachypnea, and tachycardia.

33:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).

Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

34:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

35:

Hemodynamic stability is evidenced by adequate tissue and organ perfusion. Blood pressure and heart rate are parameters commonly used to assess hemodynamic stability. Clinical findings associated with hemodynamic stability include normal peripheral pulses, warm extremities, adequate urine output, and neurological stability. The Critical level of care is appropriate for patients who develop hemodynamic instability.

36:

Despite substantial advances in treatment, sepsis remains a major cause of morbidity and mortality worldwide. Rapid

identification of patients with sepsis and early initiation of treatment have been associated with improved outcomes

(Fernando et al., Ann Intern Med 2018, 168: 266-75).

A 2016 task force convened by national societies including the Society of Critical Care Medicine (SCCM) and the European Society of Intensive Care Medicine (ESICM) proposed a new definition of sepsis (Sepsis-3). Sepsis-3 defines sepsis as life-threatening organ dysfunction caused by a dysregulated host response to infection. They introduced the quick Sequential Organ Failure Assessment (qSOFA) as a bedside screening tool for identification of sepsis, and moved away from the systemic inflammatory response syndrome (SIRS) criteria.

Nevertheless, studies indicate that the SIRS criteria appear to be significantly superior to the qSOFA for predicting sepsis. The lower sensitivity of the qSOFA raises concerns regarding potential delays in sepsis identification and treatment. A systematic review and meta-analysis of over 229,000 patients showed that while qSOFA appears to be slightly superior to SIRS in predicting in-hospital mortality, SIRS criteria were superior to qSOFA in diagnosing sepsis (Serafim et al., Chest 2018, 153: 646-55). The 2021 Surviving Sepsis Campaign Guidelines recommended against using qSOFA as a single screening tool for sepsis (Evans et al., Crit Care Med 2021, 49: e1063-e143).

37:

A Quick Sequential [Sepsis-Related] Organ Failure Assessment (qSOFA) is a bedside scoring tool composed of 3 measurements (altered mental status, tachypnea, and hypotension) that was proposed for identification of patients with sepsis. The qSOFA does not require laboratory tests and can be assessed quickly and repeatedly (Singer et al., JAMA 2016, 315: 801-10). A systematic review and meta-analysis that included over 146,000 patients found that qSOFA of at least 2 points among those with suspected infection identified patients at higher risk for mortality and ICU admission (Song et al., Crit Care 2018, 22: 28). The 2021 Surviving Sepsis Campaign Guidelines supported the role for qSOFA in identifying patients with known or suspected infection who are at risk for poor outcome, and recommended against using it as a singular screening tool for sepsis (Evans et al., Crit Care Med 2021, 49: e1063-e143).

38:

Multiple studies have shown the association between significantly elevated lactic acid levels (≥ 4 mmol/L) and increased risk of mortality in patients with sepsis and infection (Alataby et al., Med Sci Monit Basic Res 2021, 27: e927834; Baysan et al., Crit Care Explor 2020, 2: e0087; Villar et al., Infect Dis (Auckl) 2019, 12: 1178633719862776; Oedorf et al., West J Emerg Med 2017, 18: 258-66; Rhee et al., JAMA 2017, 318: 1241-9; Shapiro et al., Ann Emerg Med 2005, 45: 524-8).

39:

Lactic acidosis has been found in patients with pyelonephritis who have septic shock (e.g., sepsis with signs of organ hypoperfusion and hypotension despite adequate fluid resuscitation or needing vasoconstrictive agents) (Kozyrakakis et al., J Endourol 2020, 34: 516-22).

40:

Reference: (Dellinger et al., Crit Care Med 2013, 41: 580-637)

41:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be

delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

42:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(0.21)

1L 24%(0.24) 4L 36%(0.36)

2L 28%(0.28) 5L 40%(0.40)

3L 32%(0.32) 6L 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

43:

Volume expanders are fluids administered intravenously to increase circulatory volume. Studies have demonstrated that a balanced crystalloid solution (e.g., Lactated Ringer's) is preferable to colloids in restoration of intravascular volume in sepsis and septic shock (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Winters et al., J Emerg Med 2017, 53: 928-39). However, crystalloids have been found to be less effective than colloids at stabilizing hemodynamic endpoints in critically ill patients (Martin and Bassett, J Crit Care 2019, 50: 144-54). The type of fluid selected for administration should be based on the indication for its use and other patient-specific factors.

Instruction: In order to apply criteria for volume expanders, there should be documentation of a volume deficit supported by clinical findings. The volume of infusion is patient-specific and varies based on the cause of volume

depletion, comorbid condition, and patient response. Criteria for volume expander should not be applied for maintenance intravenous fluids or electrolyte replacement.

44:

This criteria point can be satisfied with the weight-based amount specified, or, for an average-size adult patient (~70kg), by the administration of at least 2 liters of crystalloid, a colloid equivalent (e.g., two 250 ml bottles of 5% albumin), or a combination of crystalloid and colloid (at least 1 liter of crystalloid and at least one 250 ml bottle of 5% albumin) (Ernest et al., Crit Care Med 1999, 27: 46-50).

Patients who require less than 30 ml/kg of intravenous fluids in 24 hours and do not meet other criteria for the Intermediate level of care may be reviewed at the Acute level of care; other patients who are not able to tolerate the specified amount of fluid and may require early initiation of vasopressors should be reviewed at the Critical level of care.

Since patients with sepsis at risk for worse outcomes (e.g., based on higher lactate levels) are reviewed at the Intermediate level of care, it is expected the majority will require more frequent or a higher total number of IV fluid boluses with closer monitoring and frequent reassessment. Fluid volumes in this criteria point are derived from the fluid resuscitation parameters suggested by the Surviving Sepsis Campaign for sepsis-induced hypoperfusion at a starting volume of 30 ml/kg of crystalloid (e.g., normal saline, lactated Ringers solution); the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).

Optimal amount and timing of intravenous fluids administered to septic patients remains a matter of ongoing debate. Despite a lack of randomized trials and definitive guidance on appropriate fluid resuscitation endpoints and parameters for patients thought to be at increased risk of fluid overload, such as patients with heart failure (HF) or end-stage renal disease (ESRD), observational studies suggest that septic patients with CHF and ESRD may have improved outcomes when they receive the guideline-recommended amount of IV fluid. A retrospective cohort study based on Medicare claims data showed that patients with CHF and severe sepsis or septic shock who did not receive the recommended 30 ml/kg of intravenous fluid within 3 hours of identification of sepsis had higher in-hospital and long-term mortality (Taenzer et al., Mayo Clin Proc Innov Qual Outcomes 2020, 4: 537-41). Another cohort study showed that patients with severe sepsis and septic shock who were older, had obesity, CHF, ESRD, or fluid overload, were less likely to receive 30 ml/kg of IV fluid within 3 hours; not receiving 30 ml/kg by 3 hours was associated with increased odds of in-hospital mortality and delayed hypotension (Kuttab et al., Crit Care Med 2019, 47: 1582-90). A retrospective study from Kaiser Permanente demonstrated a reduction in mortality following implementation of a 3-hour sepsis bundle which included 30 mg/kg (or 2L) of intravenous fluids, driven largely by the mortality benefit in patients with heart failure and chronic kidney disease (Liu et al., Am J Respir Crit Care Med 2016, 193: 1264-70).

45:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

46:

Acute respiratory distress syndrome (ARDS) is characterized by the presence of severe arterial hypoxemia and diffuse bilateral pulmonary infiltrates, not due exclusively to cardiogenic or hydrostatic causes. The updated Berlin definition of Acute Respiratory Distress Syndrome (ARDS) adopted in 2011 includes all of the following components (Weinberger et al., In: Weinberger et al., ed. Principles of Pulmonary Medicine, 9th edn. 2019:871-80; Force et al., JAMA 2012, 307: 2526-33):

- Timing: Within 1 week of a known clinical insult or new/worsening respiratory symptoms
- Chest imaging: Bilateral opacities-not fully explained by effusions, lobar/lung collapse, or nodules
- Origin of edema: Respiratory failure not fully explained by cardiac failure or fluid overload; need objective assessment (e.g., echocardiography) to exclude hydrostatic edema if no risk factor for ARDS is present
- Oxygenation:

Mild: 200 mm Hg < PaO₂/FiO₂ ≤ 300 mm Hg with PEEP or CPAP ≥ 5 cm H₂O (this may be delivered noninvasively in the mild ARDS group)

Moderate: 100 mm Hg < PaO₂/FiO₂ ≤ 200 mm Hg with PEEP ≥ 5 cm H₂O

Severe: PaO₂/FiO₂ ≤ 100 mm Hg with PEEP ≥ 5 cm H₂O

47:

Glasgow Coma Scale (GCS) is a tool commonly used to measure neurological function. GCS evaluates the level of consciousness in acutely ill patients. A patient's GCS is the combined total score accumulated through the evaluation of the following 3 areas (ranging from 3-15) (Chou et al., In: Glasgow Coma Scale for Field Triage of Trauma: A Systematic Review. 2017):

- Eye opening - 4 = Spontaneous, 3 = To voice, 2 = To pain, 1 = None
- Verbal response - 5 = Oriented, 4 = Confused, 3 = Inappropriate, 2 = Incomprehensible sounds, 1 = None
- Motor response - 6 = Obeys verbal commands, 5 = Localizes to pain, 4 = Withdraws to pain, 3 = Flexion response to pain, 2 = Extension response to pain, 1 = None

48:

While no randomized, controlled trials have shown that continuous renal replacement therapy (CRRT) is superior to intermittent dialysis with respect to survival, there are several advantages that may influence its use despite the lack of demonstrated survival benefit.

These advantages include:

- Gradual solute and fluid shifts resulting in greater hemodynamic stability, little risk of disequilibrium syndrome, and no worsening of brain edema
- Less need for fluid or nutritional restrictions, allowing for adequate nutrition without compromising fluid balance
- Decreased risk for intradialytic hypotension and recurrent kidney injury

Comprehensive guidelines recommend the use of CRRT for patients with hemodynamic instability, brain edema, persistent metabolic acidosis, and large fluid removal requirements. CRRT can be discontinued once renal recovery has been confirmed or another form of renal replacement is chosen based on the patient's clinical condition (Ostermann et al., Kidney Int 2020, 98: 294-309; Depret et al., Ann Intensive Care 2019, 9: 32; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii-138).

49:

Disseminated intravascular coagulation or DIC is a hematological emergency where there is abrupt systemic cascade activation of coagulation leading to thrombosis of midsize to small blood vessels rapidly leading to acute organ dysfunction. Simultaneously, there is consumption of platelets and proteins involved in the coagulation cascade, resulting in acute thrombocytopenia and low clotting factors. DIC can be caused by conditions such as sepsis, trauma, severe obstetrical complications, and malignancies. Laboratory abnormalities found in DIC include low fibrinogen (not always present), elevated fibrin degradation products, elevated D-dimer, thrombocytopenia, and prolonged prothrombin (PT) and activated partial thromboplastin times. Prompt therapy for DIC includes treatment against the underlying disease but may also include platelet transfusion, plasma (fresh frozen plasma; FFP) or factor transfusion in cases of hemorrhage, or anticoagulants in situations with thrombosis formation predominates (Wada et al., J Thromb Haemost 2013;).

50:

Critically ill patients with sepsis should be considered for a transfusion of packed red blood cells when hemoglobin is less than 7.0 g/dL (70 g/L), unless there are additional circumstances (e.g., recent or current myocardial infarction (MI), acute hemorrhage) which could potentially lead to a different transfusion threshold (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).

51:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures

fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

52:

Fluid resuscitation for patients with sepsis-induced hypoperfusion or hypotension usually begins with 30 mL per kg in adults; the recommendation for this starting amount was downgraded from a strong to a weak recommendation in the 2021 Surviving Sepsis Campaign Guidelines as a change from the 2016 Guidelines (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Semler and Rice, Clin Chest Med 2016, 37: 241-50). Traditionally, parameters used to assess response to fluid administration included improvements in mean arterial pressure (MAP), urine output, or central or mixed venous oxygen saturation. Surviving Sepsis Campaign guidelines recommend that decisions regarding additional fluid administration should be based on frequent reassessment of hemodynamic and clinical status, as well as dynamic variables of fluid responsiveness including inferior vena cava diameter, pulse pressure variation, stroke volume variation, and lactate clearance (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552).

Patients with ongoing hypotension and inability to tolerate the recommended amount of fluid due to volume overload may need early initiation of vasopressor therapy.

53:

Patients with sepsis-induced tissue hypoperfusion can present with elevated blood lactate levels greater than 2 mmol/L (18.0 mg/dL) (Shankar-Hari et al., JAMA 2016, 315: 775-87).

54:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

55:

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen (FiO_2) that is heated (up to 37 degrees Celsius) with 100% (1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the FiO_2 . The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

56:

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status

should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

57:

The initial vasopressor of choice for patients with septic shock is norepinephrine. Therapy should target a mean arterial pressure (MAP) of greater than or equal to 65 mmHg. Patients on vasoactive medication frequently have an arterial line placed for blood pressure measurement (Evans et al., Crit Care Med 2021, 49: e1063-e143; Rhodes et al., Crit Care Med 2017, 45: 486-552; Dellinger et al., Crit Care Med 2013, 41: 580-637).

58:

The 2021 Surviving Sepsis Campaign Guidelines suggest adjunct corticosteroids (typically intravenous hydrocortisone 200 mg daily in divided doses) for patients with septic shock who remain on vasopressors at a dose of 0.25 mcg/kg/min or higher of norepinephrine or epinephrine at least 4 hours after initiation (Evans et al., Crit Care Med 2021, 49: e1063-e143).

59:

Patients with pyelonephritis, or complicated urinary tract infection, requiring hospitalization should be initiated on IV antibiotics. The following parental anti-infectives are recommended (Dubbs and Sommerkamp, Emerg Med Clin North Am 2019, 37: 707-23; Gupta et al., Clin Infect Dis 2011, 52: e103-20. Reaffirmed 2013):

- Fluoroquinolone
- Aminoglycoside with or without ampicillin
- Extended spectrum cephalosporin with or without an aminoglycoside
- Carbapenem

60:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

61:

There is a role for oral antibiotics in the inpatient setting to treat the subset of patients with less severe disease whose clinical findings still support admission, but the decision to administer oral antibiotics does not drive the need for inpatient treatment.

62:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

63:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

64:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

65:

Oral intake should be at least equivalent to the patient's maintenance fluid requirements or within parameters which the medical practitioner deems appropriate. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid rates are based on ideal body weight and reflect the minimum amount of fluid required to maintain hydration. Rates may be decreased in older patients or those with a history of renal impairment or heart failure (Scales, Br J Nurs 2014, 23).

66:

Studies have demonstrated that imaging prior to 72 hours of treatment adds very little to the management of patients with pyelonephritis who are responding to treatment. If the patients do not respond to treatment by 72 hours, then imaging is recommended. Computed tomography scan of the abdomen and pelvis with and without contrast is the imaging study of choice (American College of Radiology, ACR appropriateness criteria: acute pyelonephritis. 2012. Revised 2018.).

67:

In most patients hospitalized with infection, clinical improvement would be expected to occur within 2 days. For patients who are deteriorating or not responding to treatment, it may be necessary to broaden anti-infective coverage while awaiting the results of additional diagnostic tests. In addition, consultation by an infectious disease specialist and further diagnostic tests may be valuable to evaluate reasons for the lack of improvement.

68:

Resolution of infection is indicated by a decrease in temperature, a normalizing of white blood cells and other lab values, and improving imaging studies. If there is a wound present, signs of infection should be resolving (e.g., decrease in wound drainage, less pain at site).

69:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

70:

The absolute neutrophil count (ANC) is the percentage of available polymorphonuclear cells (also referred to as polys, PMNs, or segs) and bands multiplied by the white blood cells (WBC). The resulting count predicts the body's ability to respond to an infection. Neutropenia is the condition of a low ANC. Patients with an ANC less than 500/cu. mm($500 \times 10^6/L$) are severely neutropenic and have a very limited immunologic response to infection.

To determine the ANC, a differential WBC count must be performed. Segs and bands may be reported as a percentage of the WBC or can be reported in total number.

The formula for calculating an ANC is neutrophils (segs + bands) x WBC.

Examples:

ANC calculation using a percentage of total WBC count:

WBC count: 1,500/cu.mm($1.5 \times 10^9/L$)

Segs: 20%(0.20) of the WBCs

Bands: 2%(0.02) of the WBCs

$(20\% + 2\%(0.20 + 0.02)) \times 1,500/\text{cu.mm} (1.5 \times 10^9/L)$

$= 330/\text{cu.mm} (330 \times 10^6/L)$

Normal range: 1,500-8,000/cu.mm($1.5-8.0 \times 10^9/L$)

An ANC of less than 500/cu.mm($500 \times 10^6/L$) is considered severely neutropenic.

ANC Calculation using the above numbers as whole numbers:

WBC count: 1.5

Segs: 0.2

Bands: 0.02

$(0.2 + 0.02) \times 1.5 = 0.33$

Normal range: 1.5 to 8.0

An ANC of less than 0.5 is considered severely neutropenic.

71:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including

insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.

AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.

AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., *Kidney Int* 2020, 98: 294-309; Guideline Updates, In: *Acute kidney injury: prevention, detection and management*. 2019; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii-138; Wang et al., *Am J Nephrol* 2012, 35: 349-55).

72:

In this instance, oliguria is not the result of an outlet obstruction.

73:

Urine output less than 0.5 mL/kg/h may be considered oliguria.

74:

This line of criteria refers to a patient without chronic kidney disease and an unknown creatinine baseline.

75:

For expansion of intravascular volume in patients with acute kidney injury (AKI), guidelines recommend initial treatment with an isotonic crystalloid solution (sodium chloride or sodium bicarbonate); the effects of sodium chloride on renal outcomes are being studied. Forced diuresis is not recommended because it can worsen AKI in volume-responsive patients (Ostermann et al., *Kidney Int* 2020, 98: 294-309; Guideline Updates, In: *Acute kidney injury: prevention, detection and management*. 2019; Pistolesi et al., *J Nephrol* 2018, 31: 797-812; Joannidis et al., *Intensive Care Med* 2017, 43: 730-49; *Kidney Disease: Improving Global Outcomes (KDIGO)*, *Kidney International Supplements* 2012, 2: ii-138).

76:

Anti-infective medications may cause adverse drug reactions. Symptoms of an adverse drug reaction include urticaria, angioedema, bronchospasm, pruritus, rash, vomiting, diarrhea, anaphylaxis, and thrombocytopenia.

77:

Renal abscess is a purulent infection that involves the renal parenchyma and may occur in patients with renal obstruction. Patients with pyelonephritis, who do not improve with appropriate antibiotic therapy, should be evaluated for renal abscess. Computerized tomography is the recommended diagnostic imaging. Treatment includes percutaneous drainage and supportive care with antibiotics covering gram-negative bacteria (Dubbs and Sommerkamp, *Emerg Med Clin North Am* 2019, 37: 707-23).

78:

Hyperkalemia may be associated with weakness leading to muscle twitching, paresthesia, and paralysis.

79:

Dextrose 50%(0.50) with insulin is used for the treatment of hyperkalemia. This treatment shifts potassium intracellularly, and repeated doses can be given if the hyperkalemia persists. Other treatments that may be used simultaneously include potassium binding agents, diuretics, nebulized albuterol, calcium, sodium bicarbonate, and dialysis (Depret et al., *Ann Intensive Care* 2019, 9: 32).

80:

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, *N Engl J Med* 2018, 378: 862-3).

Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zenglin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

81:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

82:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

83:

Instruction: This line of criteria is intended for the patient who has urine cultures positive for multidrug-resistant bacteria and excludes colonization.

84:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

85:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

86:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method,

discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

87:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

88:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

89:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

90:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours

- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

91:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

92:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

93:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

94:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

95:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

96:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

97:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

98:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

99:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some

physical contact to steady, guide, or move

- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

100:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

101:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

102:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

103:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

104:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.

